Figure 4.3 depicts the RQ-4 Global Hawk, a high-altitude, long-endurance UAV developed by Northrop Grumman for the US Air Force. The figure show-cases its streamlined composite airframe, long high-aspect-ratio wings, and twin V-tail configuration, which enable aerodynamic efficiency and extended endurance exceeding 36 hours. The sensor bay beneath the fuselage houses its synthetic aperture radar (SAR) and electro-optical/infrared (EO/IR) pay-loads, while the SATCOM antenna mounted on the fuselage ensures global data relay capability. The Rolls-Royce AE 3007H turbofan engine, located atop the fuselage, provides reliable propulsion for high-altitude, wide-area surveillance. This figure highlights the aircraft's design optimized for persistent, autonomous reconnaissance missions, symbolizing a milestone in third-generation UAV innovation.



Figure 4.3 RQ-4 Global Hawk (1998).

4.5.2.3 MQ-9 Reaper (2007)

The MQ-9 Reaper, developed by General Atomics Aeronautical Systems (GA-ASI) for the United States Air Force (USAF), is one of the most advanced and operationally significant UAVs of the third generation. Officially introduced in 2007, the Reaper was designed as a multi-role, hunter-killer platform capable of performing both intelligence, surveillance,